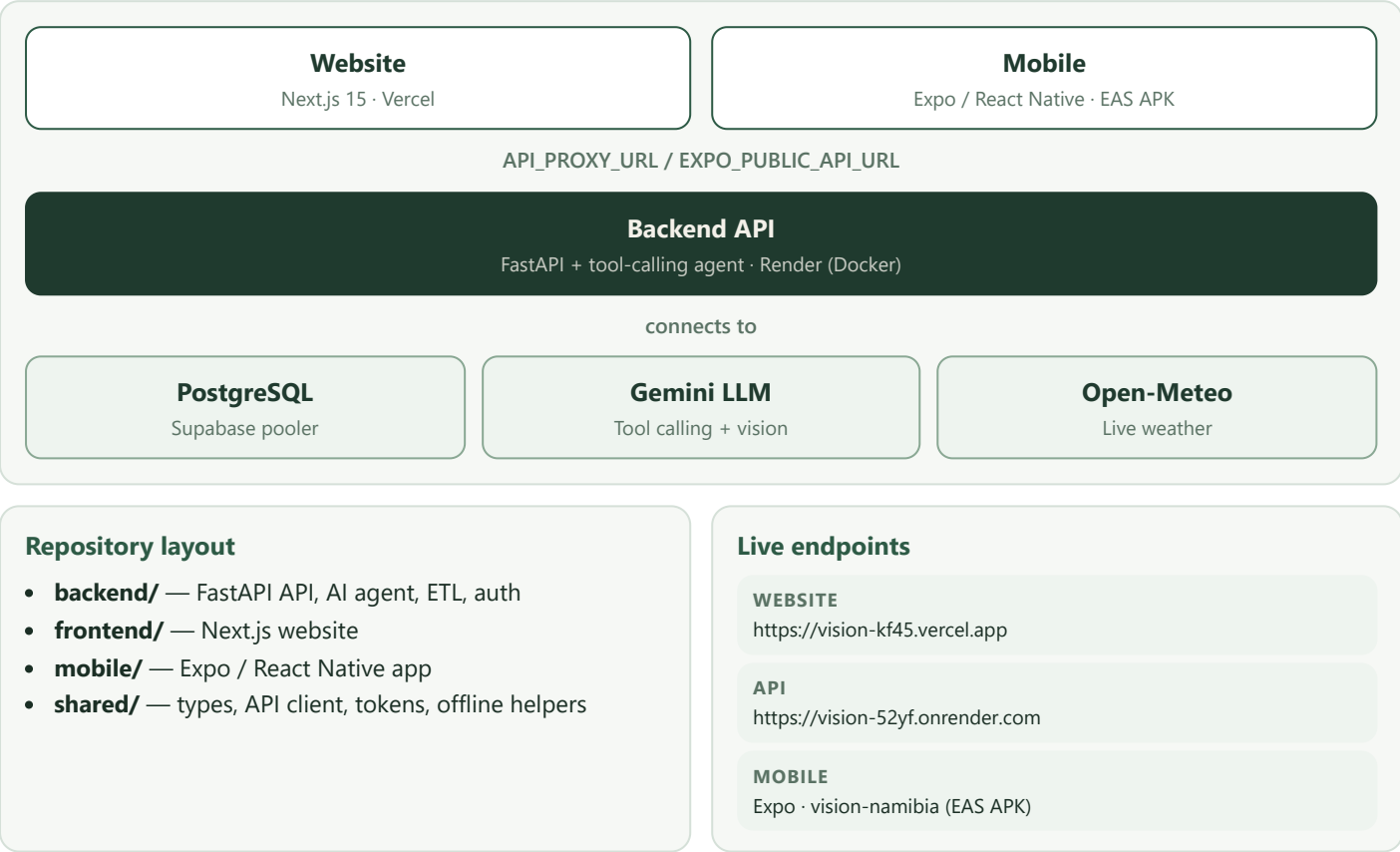


Vision

AI Rangeland & Livestock Advisor — architecture, tech stack, and deployment tools.

- Monorepo
- FastAPI + Gemini
- Next.js + Expo
- Supabase · Render · Vercel · EAS

1. System architecture



2. Assessment data flow

Farmer opens Vision → picks a camp → enters livestock → (optional photos) → runs assessment

1. Backend loads camp (farmer-provided data)
2. Fetches live Open-Meteo weather for camp coordinates
3. Searches seed data for comparable research plots
4. Computes livestock/ha (LSU) and grazing duration
5. Gemini tool-calling agent combines evidence into a structured assessment
6. Result saved to PostgreSQL with evidence, confidence, and limitations
7. Farmer can ask follow-ups in the AI advisor chat

3. Tech stack

Layer	Technologies
API	Python 3.12, FastAPI, Uvicorn, Pydantic, SQLAlchemy 2, Alembic, psycopg2, JWT + bcrypt, pandas, Pillow, httpx, OpenAI SDK
AI	Google Gemini (<code>gemini-3.1-flash-lite</code>) via OpenAI-compatible API; rule-based fallback without a key
Weather	Open-Meteo (no API key)
Database	PostgreSQL — Supabase session pooler (production); optional local Docker Postgres 16
Website	Next.js 15, React 19, TypeScript, Tailwind CSS, Leaflet, Recharts, Lucide
Mobile	Expo 52, Expo Router, React Native 0.76, Secure Store, AsyncStorage, NetInfo, Image Picker, Location, Speech
Shared	TypeScript package <code>@vision/shared</code> — API client, offline cache/queue, theme tokens

4. Deployment & tools

Role	Tool
Source control	GitHub
Database hosting	Supabase
API hosting	Render (Docker / Python 3.12)
Website hosting	Vercel (root: <code>frontend</code> , env: <code>API_PROXY_URL</code>)
Mobile builds	Expo EAS Build (Android preview APK)
Local database (optional)	Docker Compose
AI keys	Google AI Studio
Local development	Node.js 18+, Python 3.12+, Expo Go / EAS CLI

5. Backend building blocks

API modules • Auth (register / login / JWT) • Farms & camps • Assessments • Advisor chat • Camp compare • Photos & dataset	Agent & data • Tool-calling loop + prompts • Rule-based fallback engine • Reference seed → Postgres ETL • Demo farm seed • Weather & calc services
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